

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M.Pharm (Quality Assurance)

University Theory Examination December 2016

PHQA004 CLINICAL RESEARCH AND DRUG REGULATORY AFFAIRS

Date: 19.12.2016 ,Monday

Time:10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 (A) Write a note on "Para IV submission" & "Generic Drug Exclusivity Period". 05
(B) Discuss the content and format of IND application. 05
- Q.2 (A) Explain three Level of changes in Component and compositions as per SUPAC guideline 05
(B) Write a note on the types of inspections carried out as per WHO guideline. 05

OR

- Q.2 (A) Enlist different types of DMF with its contents. Explain Type I & II DMF in detail. 10
Give the difference between IND and ANDA. Why ANDA applications are called as "Abbreviated"?
- Q.3 Answer the following (ANY THREE) 15
(A) Write a note on MHRA guideline for market authorization of medicines with its type of applications.
(B) Write in detail about "Investigational Brochure".
(C) Give details of history of drug approval process.
(D) Short note on 505(b)(2) Application

SECTION-II

- Q.4 (A) What is GCP? Write ICH GCP principles. 05
(B) Describe the processes of Blinding and Randomization in Clinical Trial Design. 05
- Q.5 Give outline of Schedule Y of Drug and Cosmetic Act. 10
- OR
- Q.5 Give an overview of regulatory guidelines for medical devices in India. 10
- Q.6 Answer the following (ANY THREE) 15
(A) What are Inclusion/Exclusion Criteria? Explain them giving example.
(B) What is IRB and IEC? Write role and responsibilities of IRB/IEC.
(C) Describe any one Clinical Trial Design in detail.
(D) What is ICH? Give overview of ICH Quality Guidelines.

µg/mL is desired? Assume that the drug is given by intravenous bolus injection every 8 hours.

- D) Describe the effects of hepatic insufficiency on pharmacokinetic behavior of the drug with suitable examples.

SECTION-II

Q.4 Describe the pharmacokinetic changes in paediatric population and pregnancy. 10

Q.5 Explain the regulatory aspects of bioavailability and bioequivalence studies. 10

OR

Q.5 Discuss pharmacokinetic drug interactions with suitable examples and suggestive measures. 10

Q.6 Answer the following (Any three) 15

- (A) Describe impact of genetic variability on drug metabolism with suitable examples.
- (B) Discuss the methods used to determine bioavailability.
- (C) What is the significance of Glomerular Filtration Rate and Creatinine Clearance in the dose adjustment in patients with renal insufficiency?
- (D) Describe the historical aspects of pharmacogenetics.